ESSENTIAL OF (CS/IT/IS)

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LECTURE 3-4:

NUMBER SYSTEMS

Introduction

- A digital logic system may well have a numerical computation capability as well as its inherent logical capability.
- Human beings normally perform arithmetic operations using the decimal number system, but, a digital machine is inherently binary and its numerical calculations are executed using a binary number system.
- Since the decimal system has ten digits, a ten-state device is required to represent the decimal digits.
- Ten-state devices are not readily available in the electrical world, however two-state devices such as a transistor which are operating in a switching mode.
- A number of other systems such as the hexadecimal system are used in conjunction with programmable logic devices.

Introduction

- We are all familiar with the decimal number system (Base 10). Some other number systems that we will work with are:
  - Decimal Base 10
  - Binary Base 2
  - Octal Base 8
  - Hexadecimal Base 16

Characteristics of Numbering Systems

- The digits are consecutive.
- The number of digits is equal to the size of the base.
- Zero is always the first digit.
- The base number is never a digit.
- When 1 is added to the largest digit, a sum of zero and a carry of one results.

Significant Digits

Binary: 11101101

Most significant digit ***Least significant digit***

Hexadecimal: 1D63A7A

Most significant digit ***Least significant digit***

Binary Number System

- Also called the **"Base 2 system"**
- The binary number system is used to model the series of electrical signals computers use to represent information
- **0** represents the no voltage or an **off state**
- **1** represents the presence of voltage or an **on state**

Binary Numbering Scale

Base 2 Number	Base 10 Equivalent	Power	Positional Value
000	0	2^0	1
001	1	2^1	2
010	2	2^2	4
011	3	2^3	8
100	4	2^4	16
101	5	2^5	32
110	6	2^6	64
111	7	2^7	128

Binary Addition

4 Possible Binary Addition Combinations:

(1)

$$\begin{array}{r} 0 \\ +0 \\ \hline 00 \end{array}$$

Carry → 00 ← Sum

(2)

$$\begin{array}{r} 0 \\ +1 \\ \hline 01 \end{array}$$

(3)

$$\begin{array}{r} 1 \\ +0 \\ \hline 01 \end{array}$$

(4)

$$\begin{array}{r} 1 \\ +1 \\ \hline 10 \end{array}$$

Decimal to Binary Conversion

- The easiest way to convert a decimal number to its binary equivalent is to use the ***Division Algorithm***
- This method repeatedly divides a decimal number by 2 and records the quotient and remainder
  - ***The remainder digits (a sequence of zeros and ones) form the binary equivalent in least significant to most significant digit sequence***

Division Algorithm

Convert $(67)_{10}$ to its binary equivalent:

$67_{10} = x_2$

Step 1: 67 / 2 = 33 R 1 *Divide 67 by 2. Record quotient in next row*

Step 2: 33 / 2 = 16 R 1 *Again divide by 2; record quotient in next row*

Step 3: 16 / 2 = 8 R 0 *Repeat again*

Step 4: 8 / 2 = 4 R 0 *Repeat again*

Step 5: 4 / 2 = 2 R 0 *Repeat again*

Step 6: 2 / 2 = 1 R 0 *Repeat again*

Step 7: 1 / 2 = 0 R 1 *STOP when quotient equals 0*

$$\boxed{(1\,0\,0\,0\,0\,1\,1)_2}$$

Binary to Decimal Conversion

- The easiest method for converting a binary number to its decimal equivalent is to use the ***Multiplication Algorithm***
- Multiply the binary digits by increasing powers of two, starting from the right
- Then, to find the decimal number equivalent, sum those products

Multiplication Algorithm

Convert $(10101101)_2$ to its decimal equivalent:

Binary	1	0	1	0	1	1	0	1	
	x	x	x	x	x	x	x	x	
Positional Values	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
Products	128 + 32 + 8 + 4 + 1								

$$\boxed{(173)_{10}}$$

Octal Number System

- Also known as the Base 8 System
- Uses digits 0 - 7
- Readily converts to binary
- Groups of three (binary) digits can be used to represent each octal digit
- Also uses multiplication and division algorithms for conversion to and from base 10

Decimal to Octal Conversion

Convert $(427)_{10}$ to its octal equivalent:

427 / 8 = 53 R3	Divide by 8; R is LSD
53 / 8 = 6 R5	Divide Q by 8; R is next digit
6 / 8 = 0 R6	Repeat until Q = 0

$$\boxed{(653)_8}$$

Octal to Decimal Conversion

Convert $(653)_8$ to its decimal equivalent:

Octal Digits → 6 5 3

x x x

Positional Values → 8^2 8^1 8^0

Products → 384 + 40 + 3

$$\boxed{(427)_{10}}$$

Octal to Binary Conversion

Each octal number converts to 3 binary digits

Code
0 - 000
1 - 001
2 - 010
3 - 011
4 - 100
5 - 101
6 - 110
7 - 111

To convert $(653)_8$ to binary, just substitute code:

6	5	3
↓	↓	↓
110	101	011

Hexadecimal Number System

- Base 16 system
- Uses digits 0-9 & letters A,B,C,D,E,F
- Groups of four bits represent each base 16 digit

Decimal	Hexadecimal
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

Decimal to Hexadecimal Conversion

Convert $(830)_{10}$ to its hexadecimal equivalent:

830 / 16 = 51 R14 ← = E in Hex

51 / 16 = 3 R3

3 / 16 = 0 R3

$\boxed{(33E)_{16}}$

Hexadecimal to Decimal Conversion

Convert $(3B4F)_{16}$ to its decimal equivalent:

Hex Digits	3	B	4	F
	x	x	x	x
Positional Values	16^3	16^2	16^1	16^0
Products	12288	+2816	+ 64	+15

$$\boxed{(15,183)_{10}}$$

Binary to Hexadecimal Conversion

- The easiest method for converting binary to hexadecimal is to use a substitution code
- Each hex number converts to 4 binary digits

Substitution Code

0000 = 0	0100 = 4	1000 = 8	1100 = C
0001 = 1	0101 = 5	1001 = 9	1101 = D
0010 = 2	0110 = 6	1010 = A	1110 = E
0011 = 3	0111 = 7	1011 = B	1111 = F

Substitution Code

Convert $(010101101010111001101010)_2$ to hex using the 4-bit substitution code :

0101 0110 1010 1110 0110 1010

5 6 A E 6 A

$$\boxed{(56AE6A)_{16}}$$

Substitution Code

Substitution code can also be used to convert binary to octal by using 3-bit groupings:

010 101 101 010 111 001 101 010

2 5 5 2 7 1 5 2

$$\boxed{(25527152)_8}$$

Complementary Arithmetic

- 1's complement
  - Switch all 0's to 1's and 1's to 0's

Binary #	10110011
1's complement	01001100

- The 2's complement representation of a binary number *X* is defined by the equation:

$$[X]_2 = 2^n - X$$

For X = 1010 and n = 4 then:

$$[X]_2 = 2^4 - 1010$$

$$= 10000 - 1010 = 0110$$

Complementary Arithmetic

- 2's complement
  - Step 1: Find 1's complement of the number

    Binary # 11000110

    1's complement 00111001
  - Step 2: Add 1 to the 1's complement

$$
\begin{array}{r}
00111001 \\
+ \quad 1 \\
\hline
00111010
\end{array}
$$

Signed Numbers

Need to be able to represent both *positive* and *negative* numbers
- Following 3 representations

Signed magnitude representation Signed 1's complement representation Signed 2's complement representation

Example: Represent (+9) and (-9) in 8 bit-binary number

Only one way to represent +9 ==> 0 0001001
Three different ways to represent -9:
In signed-magnitude: 1 0001001
In signed-1's complement: 1 1110110
In signed-2's complement: 1 1110111

Signed Numbers

Complement:

Signed magnitude: Complement ***only*** the sign bit
Signed 1's complement: Complement ***all*** the bits including sign bit
Signed 2's complement: After the first (1) of the positive number complement the bits ***including*** its sign bit.

Singed Binary Numbers

- In computers, both positive and negative numbers are represents only with binary digits;
- The left most bit (*sign bit*) in the number represents sign of the number;
- The sign bit is **0** for the positive numbers, and **1** for negative numbers.

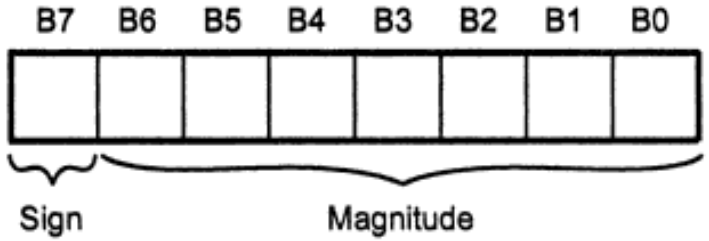


Fig. 1.3

1) + 6 = 0000 0110
2) – 14 = 1000 1110
3) + 24 = 0001 1000
4) – 64 = 1100 0000

$(-23)_{10} = (1, 0010111)_2$ $\quad$ $(+23)_{10} = (0, 0010111)_2$

$(-0)_{10} = (1, 0000000)_2$ $\quad$ $(+0)_{10} = (0, 0000000)_2$

Singed Binary Numbers

Addition and subtraction of 2's complement numbers:

Addition and subtraction in the 2's complement system are both carried out as **additions**.

Example 1: Addition of two 8-bit positive numbers.

	-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
	0,	0	1	0	1	1	0	0	+44
+	0,	0	1	1	0	1	0	0	+52
	0,	1	1	0	0	0	0	0	+96

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
0,	1	1	0	0	0	1	1	+99
1,	1	0	1	1	1	1	1	−33
0,	1	0	0	0	0	1	0	66

Operations on Binary system

The rules for the **addition** of two single-bit numbers as following :

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

```
Carry ---->  1  1  1
             1  0  1  1
             0  1  1  1
          ----------------
          1  0  0  1  0
Sum
```

```
   11
 + 7
 ----
   18
```

Operations on Binary system

The rules for the **subtraction** of two single-bit numbers as following :

A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Borrow →

```
   1 1
 1 1 0 0
 0 0 1 1
 -------
 1 0 0 1
```

Deff.

```
  12
- 3
----
   9
```

Operations on Binary system

The rules for the **Multiplication** of two single-bit numbers as following :

A	B	Product
0	0	0
0	1	0
1	0	0
1	1	1

```
     1110
   x 1011
  ------
     1110
    1110
   0000
  1110
  ------
10011010
```

Conversion of decimal numbers to any Radix number

Conversion of decimal to any radix number accomplished in two steps:

Step 1: Convert the ***Integer*** part – by successive division method;

Step 2: Convert the ***Fraction*** part - by successive multiplication method.

Conversion of decimal numbers to any Radix number

1. successive division for integer part:

Example 1.20 : *Convert decimal number 37 to its binary equivalent.*

Solution : Here r is 2

		R	
Divide 37 by 2	$2\overline{)37}$ = 18, − 36, 1	1	LSD
Divide 18 by 2	$2\overline{)18}$ = 9, − 18, 0	0	
Divide 9 by 2	$2\overline{)9}$ = 4, − 8, 1	1	
Divide 4 by 2	$2\overline{)4}$ = 2, − 4, 0	0	
Divide 2 by 2	$2\overline{)2}$ = 1, − 2, 0	0	
Divide 1 by 2	$2\overline{)1}$ = 0, − 0, 1	1	MSD

≡

	Q	R	
2	37	1	LSD
2	18	0	
2	9	1	
2	4	0	
2	2	0	
2	1	1	MSD
	0		

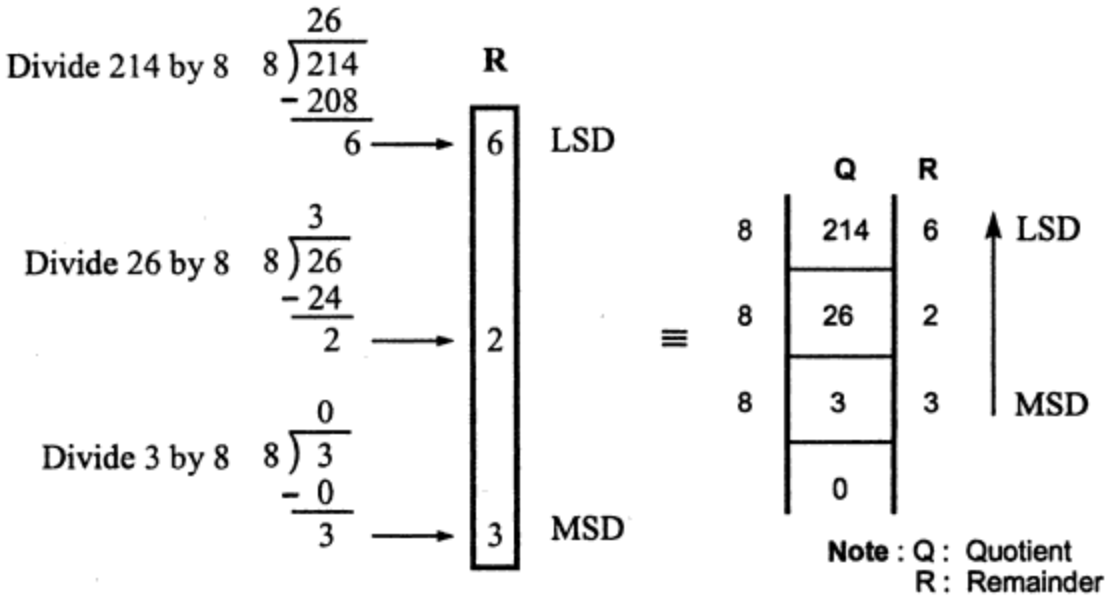
Note : Q : Quotient
R : Remainder

Binary equivalent = 100101_2

Conversion of decimal numbers to any Radix number

Example 1.21 : *Convert decimal number 214 to its octal equivalent.*

Solution : Here r is 8



The conversion is over when quotient is 0, and we get 326_8 as octal equivalent to decimal number 214.

Conversion of decimal numbers to any Radix number

2. successive multiplication for Fraction part:

Example 1.24 : *Convert 0.8125 decimal number to its binary equivalent.*

Solution :

Fraction		Radix		Result			Recorded carries	
0.8125	×	2	=	1.625	=	0.625	with a carry of 1	MSD
0.625	×	2	=	1.25	=	0.25	with a carry of 1	↓
0.25	×	2	=	0.5	=	0.5	with a carry of 0	
0.5	×	2	=	1.0	=	0.0	with a carry of 1	LSD

Reading carries downward we get,

Binary fraction = 0.1101, which is equivalent to 0.8125 decimal.

Example 1.26 : *Convert 0.640625 decimal number to its octal equivalent.*

Solution :

Fraction		Radix		Result			Recorded carries	
0.640625	×	8	=	5.125	=	0.125	with a carry of 5	MSD
0.125	×	8	=	1.0	=	0	with a carry of 1	LSD

Reading carries downward we get octal fraction = 0.51, which is equivalent to 0.640625 decimal.

Conversion of decimal numbers to any Radix number

Example 1.28 : *Convert decimal number 24.6 to a binary number.*

Solution :

Step 1 : Separate out integer part and fractional part.

Integer part : 24 Fractional part : 0.6

Step 2 : Find equivalent binary number for integer part

		R	
Divide 24 by 2	$2\overline{)24}$ = 12, -24, 0	0	LSD
Divide 12 by 2	$2\overline{)12}$ = 6, -12, 0	0	
Divide 6 by 2	$2\overline{)6}$ = 3, -6, 0	0	
Divide 3 by 2	$2\overline{)3}$ = 1, -2, 1	1	
Divide 1 by 2	$2\overline{)1}$ = 0, -0, 1	1	MSD

≡

	Q	R	
2	24	0	LSD
2	12	0	
2	6	0	
2	3	1	
2	1	1	MSD
	0		

Note : Q : Quotient
R : Remainder

Binary equivalent of integer part = 11000_2

Fractions

Step 3 : Find equivalent binary number for fractional part

Fraction		Radix		Result			Recorded carries	
0.6	×	2	=	1.2	=	0.2	with a carry of 1	MSD
0.2	×	2	=	0.4	=	0.4	with a carry of 0	↓
0.4	×	2	=	0.8	=	0.8	with a carry of 0	
0.8	×	2	=	1.6	=	0.6	with a carry of 1	
0.6	×	2	=	1.2	=	0.2	with a carry of 1	LSD

The binary equivalent number is 11000.10011. This number is an approximation of decimal 24.6, because we have terminated the conversion of fractional part after 5 digits.

Fractions

Convert Decimal to Binary

- $(3.14579)_{10} = (?)_2$

```
3.14579

11.001001...
```

```
 .14579
x      2
0.29158
x      2
0.58316
x      2
1.16632
x      2
0.33264
x      2
0.66528
x      2
1.33056
etc.
```

Operation in Octal

The rules for the **Addition** of two numbers as following :

+	٠	١	٢	٣	٤	٥	٦	٧
٠	٠	١	٢	٣	٤	٥	٦	٧
١	١	٢	٣	٤	٥	٦	٧	١٠
٢	٢	٣	٤	٥	٦	٧	١٠	١١
٣	٣	٤	٥	٦	٧	١٠	١١	١٢
٤	٤	٥	٦	٧	١٠	١١	١٢	١٣
٥	٥	٦	٧	١٠	١١	١٢	١٣	١٤
٦	٦	٧	١٠	١١	١٢	١٣	١٤	١٥
٧	٧	١٠	١١	١٢	١٣	١٤	١٥	١٦

$$\begin{array}{r} (3\ \ 4)_8 \\ +\ (4\ \ 2)_8 \\ (7\ \ 6)_8 \\ \hline \end{array}$$

$$\begin{array}{r} (5\ \ 6)_8 \\ +\ (6\ \ 3)_8 \\ (1\ 4\ 1)_8 \\ \hline \end{array}$$

Operation in Octal

The sum of two octal digits is the same as their decimal sum, provided the decimal sum is less than 8. If the decimal sum is 8 or greater, subtract 8 to obtain the octal digit. A carry of 1 is produced when the decimal sum is corrected this way, as illustrated in the following examples.

Example 1.66 : *a)* $4_8 + 2_8$ *b)* $6_8 + 7_8$ *c)* $1_8 + 7_8$.

Solution : a) $4_8 + 2_8 = 6_8$

b) $6_8 + 7_8 = (13 - 8)\ 5_8$ carry 1

c) $1_8 + 7_8 = (8 - 8)\ 0_8$ carry 1

Note : Subscript 8 denotes it is an octal number

To obtain the sum of multi-digit octal numbers, the procedure just described is applied to each column of digits as illustrated in the following example.

Example 1.67 : *Add* 167_8 *and* 325_8.

Solution :

```
  1   1          ← Carry
  1   6   7
+ 3   2   5
-----------
  5   1   4₈
```

Operation in Octal

Subtraction with 7's complement

The 7's complement of an octal number is found by subtracting each digit from 7, as illustrated in Ex. 1.69.

Example 1.69 : *Find 7's complement of* 612_8.

Solution :

$$\begin{array}{rccc} & 7 & 7 & 7 \\ - & 6 & 1 & 2 \\ \hline & 1 & 6 & 5_8 \end{array}$$

The steps for octal subtraction using 7's complement method are as given below :

Step 1 : Find 7's complement of subtrahend

Step 2 : Add two octal numbers (first number and 7's complement of the second number)

Step 3 : If carry is produced in the addition, add carry in the least significant bit of the sum; otherwise find 7's complement of the sum as a result with negative sign.

Operation in Octal

Example 1.70 : *Use the 7's complement method of subtraction to compute* $176_8 - 157_8$.

Solution :

Step 1 :

$$\begin{array}{r} 7\ 7\ 7 \\ -\ 1\ 5\ 7 \\ \hline 6\ 2\ 0 \end{array}$$

7's complement

Step 2 :

$$\begin{array}{rr} & \mathbf{1} \\ \leftarrow \text{carry} & 1\ 7\ 6 \\ + & 6\ 2\ 0 \\ \hline \text{carry} \rightarrow & 1\ 0\ 1\ 6 \end{array}$$

Step 3 :

$$\begin{array}{r} 0\ 1\ 6 \\ +\ \ \ \ \ 1 \\ \hline 0\ 1\ 7 \end{array}$$

$\therefore \quad 176_8 - 157_8 \rightarrow 017_8$

Operation in Octal

Example 1.71 : *Use the 7's complement method of subtraction to compute* $153_8 - 243_8$.

Solution :

Step 1 :

$$\begin{array}{rccc} & 7 & 7 & 7 \\ - & 2 & 4 & 3 \\ \hline & 5 & 3 & 4 \\ & 1 & & \end{array}$$

← 7's complement

← carry

Step 2 :

$$\begin{array}{rccc} & 1 & 5 & 3 \\ + & 5 & 3 & 4 \\ \hline & 7 & 0 & 7 \end{array}$$

← 7's complement

Step 3 : No carry, hence take 7's complement

$$\begin{array}{rccc} & 7 & 7 & 7 \\ - & 7 & 0 & 7 \\ \hline & 0 & 7 & 0 \end{array}$$

← 7's complement of sum (– ve sum)

$\therefore \quad 153_8 - 243_8 \rightarrow 70_8$

Operation in Octal

Subtraction with 8's Complement

The 8's complement of an octal number is found by adding a 1 to the least significant bit of the 7's complement of an octal number, as illustrated in the Ex. 1.71.

Example 1.72 : *Find the 8's complement of* 346_8.

Solution :

$$
\begin{array}{rccc l}
 & 7 & 7 & 7 & \\
- & 3 & 4 & 6 & \\
\hline
 & 4 & 3 & 1 & \leftarrow \text{7's complement} \\
+ & 0 & 0 & 1 & \leftarrow \text{Add 1} \\
\hline
 & 4 & 3 & 2 & \leftarrow \text{8's complement}
\end{array}
$$

The steps for octal subtraction using 8's complement method are as given below :

Step 1 : Find 8's complement of subtrahend

Step 2 : Add two octal numbers (first number and 8's complement of the second number).

Step 3 : If carry is produced in the addition it is discarded ; otherwise find 8's complement of the sum as a result with negative sign.

Example 1.73 : *Use the 8's complement method of subtraction to compute* $516_8 - 413_8$.

Operation in Octal

Solution :

Step 1 :

	7	7	7	
−	4	1	3	
	3	6	4	← 7's complement
			1	← Add 1
	3	6	5	← 8's complement
		1	1	← carry

Step 2 :

	5	1	6	
+	3	6	5	
carry → 1	1	0	3	

Step 3 : Carry is ignored

$\therefore \quad 516_8 - 413_8 \rightarrow 103_8$

Operation in Hexa

The rules for the **Addition** of two numbers as following :

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	+
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0	0
10	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	1
11	10	F	E	D	C	B	A	9	8	7	6	5	4	3	2	2
12	11	10	F	E	D	C	B	A	9	8	7	6	5	4	3	3
13	12	11	10	F	E	D	C	B	A	9	8	7	6	5	4	4
14	13	12	11	10	F	E	D	C	B	A	9	8	7	6	5	5
15	14	13	12	11	10	F	E	D	C	B	A	9	8	7	6	6
16	15	14	13	12	11	10	F	E	D	C	B	A	9	8	7	7
17	16	15	14	13	12	11	10	F	E	D	C	B	A	9	8	8
18	17	16	15	14	13	12	11	10	F	E	D	C	B	A	9	9
19	18	17	16	15	14	13	12	11	10	F	E	D	C	B	A	A
1A	19	18	17	16	15	14	13	12	11	10	F	E	D	C	B	B
1B	1A	19	18	17	16	15	14	13	12	11	10	F	E	D	C	C
1C	1B	1A	19	18	17	16	15	14	13	12	11	10	F	E	D	D
1D	1C	1B	1A	19	18	17	16	15	14	13	12	11	10	F	E	E
1E	1D	1C	1B	1A	19	18	17	16	15	14	13	12	11	10	F	F

carry

1 1 1

$(3\ 5\ A\ B\ 2)_{16}$

$(1\ A\ 6\ 7\ 5)_{16}$

$(5\ 0\ 1\ 2\ 7)_{16}$

Operation in Hexa

1.10 Hexadecimal Arithmetic

Answer following question after reading this topic.

1. *Perform addition or subtraction of any two hexadecimal numbers.*

Most likely and **Important** Question

1.10.1 Addition

The sum of two hexadecimal digits is the same as their equivalent decimal sum (with any sum from 10 through is expresses a letter from A through F), provided the decimal equivalent is less than 16. If the decimal sum is 16 or greater, subtract 16 to obtain the hexadecimal digit. A carry of 1 is produced when the decimal sum is corrected this way, as illustrated in the following examples.

Example 1.75 : *a)* $3_{16} + 9_{16}$ *b)* $9_{16} + 7_{16}$ *c)* $A_{16} + 8_{16}$.

Solution : a) $3_{16} + 9_{16} = C_{16}$

b) $9_{16} + 7_{16} = (16 - 16) \quad 0_{16}$ carry 1

c) $A_{16} + 8_{16} = (18 - 16) \quad 2_{16}$ carry 1

Operation in Hexa

To obtain the sum of multi-digit hexadecimal numbers, the procedure just described is applied to each column of digits as illustrated in the following example.

Example 1.76 : *Add $3F8_{16}$ and $5B3_{16}$.*

Solution :

1			← Carry
3	F	8	
5	B	3	
8	(26 – 16)	B	

1			← Carry
3	F	8	
5	B	3	
9	A	B_{16}	

If the decimal sum of several hex digits is 32 or greater, subtract 32 and set carry equal to 2. In general, we can expresss any decimal sum in hexadecimal form by repeatedly subtracting 16 until the result is one of the hex digits through 0 to F. Each time 16 is subtracted the amount of the carry is increased by 1.

Hexadecimal subtraction is best accomplished using the complement method. The 15's and 16's complements for hex numbers are found and used like 1's and 2's complements to perform subtraction.

Operation in Hexa

Subtraction with 15's Complement

The 15's complement of a hexadecimal number is found by subtracting each digit from 15, as illustrated in Ex. 1.77.

Example 1.77 : *Find 15's complement of $A9B_{16}$.*

Solution :

$$\begin{array}{rccc} & 15 & 15 & 15 \\ - & A & 9 & B \\ \hline & 5 & 6 & 4_{16} \end{array}$$

The steps for hexadecimal subtraction using 15's complement method are as given below :

Step 1 : Find 15's complement of subtrahend.

Step 2 : Add two hexadecimal numbers (first number and 15's complement of the second number.)

Step 3 : If carry is produced in the addition, add carry to the least significant bit of the sum; otherwise find 15's complement of the sum as a result with a negative

Operation in Hexa

Example 1.78 : *Use the 15's complement method of subtraction to compu*
$B02_{16} - 98F_{16}$.

Solution :

Step 1:

$$\begin{array}{rrrr} & 15 & 15 & 15 \\ - & 9 & 8 & F \\ \hline & 6 & 7 & 0 \end{array} \quad \leftarrow \text{15's complement}$$

Step 2 :

$$\begin{array}{rrrrr} & & B & 0 & 2 \\ & + & 6 & 7 & 0 \\ \hline \text{Carry} \rightarrow & \boxed{1} & 1 & 7 & 2 \end{array}$$

Step 3 :

$$\begin{array}{rrrr} & 1 & 7 & 2 \\ + & & & 1 \\ \hline & 1 & 7 & 3 \end{array}$$

Operation in Hexa

$\therefore \quad B\,0\,2_{16} - 9\,8\,F_{16} \rightarrow 1\,7\,3_{16}$

Example 1.79 : *Use the 15's complement method of subtraction to compute* $69B_{16} - C14_{16}$.

Solution :

Step 1 :

$$\begin{array}{rccc} & 15 & 15 & 15 \\ - & C & 1 & 4 \\ \hline & 3 & E & B \end{array} \leftarrow \text{15's complement}$$

Step 2 :

$$\begin{array}{rccc} & \boxed{1} & \boxed{1} & \\ & 6 & 9 & B \\ + & 3 & E & B \\ \hline & A & 8 & 6 \end{array}$$

Operation in Hexa

Step 3 : No carry, hence take 15's complement

$$\begin{array}{rccc} & 15 & 15 & 15 \\ - & A & 8 & 6 \\ \hline & 5 & 7 & 9 \end{array} \leftarrow \text{15's complement of sum}$$

$\therefore\ 69B_{16} - C14_{16} \rightarrow -579_{16}$

Subtraction with 16's complement

The 16's complement of a hexadecimal number is found by adding a 1 to the least significant bit of the 15's complement of a hexadecimal number, as illustrated in the Ex. 1.80.

Operation in Hexa

Example 1.80 : *Find the 16's complement of $A8C_{16}$.*

Solution :

	15	15	15	
−	A	8	C	
	5	7	3	← 15's complement
+	0	0	1	← Add 1
	5	7	4	← 16's complement

The steps for hexadecimal subtraction using 16's complement method are as given below :

Step 1 : Find 16's complement of subtrahend.

Step 2 : Add two hexadecimal numbers (first number and 16's complement of the second number).

Step 3 : If carry is produced in the addition it is discarded; otherwise find 16's complement of the sum as a result with negative sign.

Operation in Hexa

Example 1.81 : *Use the 16's complement method of subtraction to compute* $CB2_{16} - 972_{16}$.

Solution :

Step 1 :

	15	15	15	
−	9	7	2	
	6	8	D	← 15's complement
+			1	← Add 1
	6	8	E	← 16's complement

	1	1		← Carry

Step 2 :

	C	B	2	
+	6	8	E	
Carry →	1 3	4	0	

Step 3 : Carry is ignored

$\therefore CB2_{16} - 972_{16} \rightarrow 340_{16}$

Operation in Hexa

Example 1.82 : *Use the 16's complement method of subtraction to compute* $3B7_{16} - 854_{16}$

Solution :

Step 1:

	15	15	15	
−	8	5	4	
	7	A	B	← 15's complement
+	0	0	1	← Add 1
	7	A	C	← 16's complement

Step 2 :

	1	1	
	3	B	7
+	7	A	C
	B	6	3

Step 3 : No carry, hence take 16's complement

	15	15	15	
−	B	6	3	
	4	9	C	← 15's complement
+	0	0	1	← Add 1
	4	9	D	← 16's complement of sum

$\therefore \quad 3B7_{16} - 854_{16} \rightarrow -49D_{16}$